

NORSEMAN AP, AustraliaWMO: **956420**Lat: **32.2147S**Lon: **121.7547E**Elev: **863**StdP: **14.24**Time Zone: **8.00 (E08)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.6	33.9	24.7	19.4	60.3	27.0	21.6	57.4	26.9	60.7	24.9	60.7	2.2	210	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.6	101.1	66.1	97.3	65.8	93.6	64.8	71.3	87.8	69.3	86.8	67.8	84.6	14.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	102.4	73.8	64.9	95.3	72.8	63.0	88.9	72.2	35.7	87.1	33.9	86.7	32.6	85.2	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	23.1	20.2	26.8	108.3	1.9	3.0	25.4	110.4	24.3	112.1	23.2	113.8	21.8	115.9		
(5)			25.9	74.9	1.9	2.4	24.5	76.7	23.4	78.1	22.4	79.4	21.0	81.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	75.0	74.4	70.2	64.3	56.6	52.1	50.5	52.9	56.8	63.4	68.2	72.0	(6)	(7)
	DBStd	10.65	6.54	6.69	6.82	6.38	5.51	4.61	4.45	5.40	6.34	6.96	7.29	7.31	(7)	(8)
	HDD50	121	0	0	0	0	6	28	48	28	10	1	0	0	(8)	(9)
	HDD65	2040	3	6	26	88	268	388	450	378	256	115	47	15	(9)	(10)
	CDD50	4854	774	682	626	430	211	90	64	117	214	417	547	682	(10)	(11)
	CDD65	1299	312	268	187	68	8	0	0	2	11	66	144	233	(11)	(12)
	CDH74	16120	3821	3098	2050	673	93	4	1	66	275	1015	1910	3114	(12)	(13)
	CDH80	7839	2065	1591	936	206	17	0	0	14	80	401	880	1649	(13)	(14)
(14) Wind	WSAvg	9.6	10.8	10.4	9.7	8.5	8.0	8.3	8.4	9.0	9.8	10.6	10.9	10.9	(14)	(15)
(15) Precipitation	PrecAvg	12.0	0.9	1.1	1.1	1.0	1.3	1.3	1.2	0.9	0.9	0.7	0.9	0.9	(15)	(16)
	PrecMax	24.0	4.6	6.3	7.4	3.6	4.4	4.1	3.1	2.6	2.6	2.8	3.4	5.9	(16)	(17)
	PrecMin	7.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.2	0.0	0.0	0.0	0.0	(17)	(18)
	PrecStd	3.4	1.1	1.4	1.6	0.8	1.1	0.8	0.7	0.6	0.7	0.6	0.7	1.1	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	106.1	104.2	100.6	91.3	82.5	72.7	71.6	82.1	87.3	94.0	100.0	104.3	(19)	(20)
		MCWB	67.7	67.0	64.8	64.2	60.5	55.3	53.9	56.0	58.2	60.9	63.1	65.3	(20)	(21)
	2%	DB	100.4	98.8	94.7	85.7	76.0	68.3	67.6	74.2	81.2	89.2	94.2	99.5	(21)	(22)
		MCWB	66.5	67.8	64.5	62.0	57.9	54.5	52.9	54.5	57.2	60.4	63.2	65.5	(22)	(23)
	5%	DB	96.1	94.4	89.5	81.2	72.0	65.2	64.6	69.6	76.4	84.6	89.4	94.5	(23)	(24)
		MCWB	66.4	67.0	64.3	61.4	56.8	53.4	52.5	53.5	56.2	59.5	62.1	64.6	(24)	(25)
	10%	DB	91.1	89.8	84.9	77.2	68.8	62.6	61.7	65.6	72.0	79.6	84.8	89.4	(25)	(26)
		MCWB	65.5	65.9	63.5	60.2	56.0	52.5	51.5	52.5	54.7	58.4	60.6	63.3	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.5	74.1	69.7	67.5	64.0	59.3	57.9	59.3	61.8	65.4	68.1	71.5	(27)	(28)
		MCDB	89.0	86.5	83.4	80.6	74.3	67.4	63.9	70.1	77.8	81.7	85.2	88.7	(28)	(29)
	2%	WB	71.7	71.5	67.9	65.2	61.1	57.2	55.5	57.3	59.2	62.7	66.3	68.5	(29)	(30)
		MCDB	87.2	88.7	82.6	76.7	69.7	63.9	62.8	66.9	75.1	80.8	84.4	88.9	(30)	(31)
	5%	WB	69.4	69.3	66.5	63.6	59.1	55.4	53.9	55.4	57.4	61.0	64.4	66.3	(31)	(32)
		MCDB	87.3	86.5	80.3	74.7	67.7	61.7	61.2	66.8	72.3	78.9	81.3	87.4	(32)	(33)
	10%	WB	67.4	67.7	65.0	62.1	57.3	53.8	52.4	53.6	55.6	59.4	62.5	64.3	(33)	(34)
		MCDB	84.2	84.1	79.3	73.6	66.6	60.4	59.9	63.9	69.9	76.4	80.7	85.2	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	27.6	26.0	24.6	24.3	23.8	21.7	22.4	24.9	27.4	26.8	27.0	28.4	(35)	(36)
		MCDBR	35.2	32.7	31.5	28.9	28.7	25.8	27.3	31.8	34.2	34.9	33.7	35.6	(36)	(37)
	5% WB	MCWBR	10.5	9.2	9.8	10.9	14.0	14.5	15.3	16.1	14.7	12.7	10.7	10.6	(37)	(38)
		MCWBR	28.6	27.3	24.4	22.5	22.2	20.4	21.9	25.4	29.0	28.4	28.9	31.4	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.338	0.338	0.320	0.309	0.293	0.281	0.280	0.288	0.307	0.319	0.326	0.326	(39)	(40)
		taud	2.591	2.603	2.637	2.652	2.667	2.679	2.683	2.639	2.565	2.555	2.564	2.596	(40)	(41)
	Ebn at Noon		318	311	307	294	284	280	286	297	306	314	319	322	(41)	(42)
		Edh at Noon	33	32	29	25	22	21	22	25	30	33	34	33	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		2418	2126	1778	1358	1026	874	943	1242	1662	2031	2305	2518	(43)	(44)
	RadStd		147	171	118	78	62	43	52	63	66	127	145	155	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-112	+157	+72

Nomenclature: See separate page